

Module - 2

Module-2(5 Hours)

Literature Review and Technical Reading, New and Existing Knowledge, Analysis and Synthesis of Prior Art Bibliographic Databases, Web of Science, Google and Google Scholar, Effective Search: The Way Forward Introduction to Technical Reading Conceptualizing Research, Critical and Creative Reading, Taking Notes While Reading, Reading Mathematics and Algorithms, Reading a Datasheet.

Attributions and Citations: Giving Credit Wherever Due, Citations: Functions and Attributes, Impact of Title and Keywords on Citations, Knowledge Flow through Citation, Citing Datasets, Styles for Citations, Acknowledgments and Attributions, What Should Be Acknowledged, Acknowledgments in, Books Dissertations, Dedication or Acknowledgments.

- The main goal of a literature review is to **explore existing ideas and methods in order to clearly define a problem** that may not be fully understood at first. It helps the researcher choose the best way to study the problem and decide on the methods to use.
- A good literature review also **shows** the researcher **how their work will add something new and original to the field**.
- The quality of the review can be judged by **how well it covers the topic, whether it's clear and detailed, and if the analysis is thorough and consistent**.



NEW AND EXISTING KNOWLEDGE

- New knowledge in research **builds on what is already known**. It **cannot exist without the foundation of existing knowledge**.
- The **interpretation of new knowledge can vary depending on the researcher's background**, and their **reaction to it may shift from indifference to excitement based on their understanding**.
- **To show that new knowledge is important**, you need to **prove that a real problem exists**, which is **often revealed by studying what's already known in the field**.

Example: If you're researching a new way to recycle plastic, you first need to highlight the problem of plastic pollution. This helps show why your research is needed.

- **Existing knowledge is needed to explain why a problem matters and why it should be studied.**

Example: If studies show that plastic waste harms marine life, this existing knowledge helps prove that reducing plastic use is important for protecting the environment.

- To show that your research is original, you must explain what is already known and point out what is still missing. You learn this by reviewing both older knowledge and recent updates in your field, as knowledge is always changing.

Example: If previous studies have focused on recycling bottles but not bags, you can argue that your research on recycling plastic bags is original and needed.

- Older, well-established knowledge is often found in textbooks, while newer research can be found in research papers.
- Research papers are written for experts in the field, assuming the readers already have a deep understanding of the subject.
- A **good literature review connects different pieces of research**, helping knowledge grow, eliminating irrelevant areas, and opening up new paths for future studies.

- Doing a **thorough literature review** is often the **first task expected** by a **research supervisor**, and **doing it well shows a strong grasp of the subject**.
- A **good review doesn't jump to conclusions**. Instead, it carefully examines each source to understand its key ideas and combine them in a meaningful way.

Example:

Suppose you are conducting a literature review on the effectiveness of different teaching methods in online education. Rather than concluding early that one method (e.g., video lectures) is the best, a good review would involve carefully analyzing multiple studies.

For instance, one study might show that video lectures improve student engagement, while another suggests that interactive quizzes enhance knowledge retention. Instead of favoring one method over the other, you would explore the key ideas from both studies and combine them to suggest that a combination of video lectures and interactive quizzes might provide a more comprehensive learning experience. This approach leads to a more thoughtful and meaningful conclusion.

- A **good literature survey** is typically a two-step process as enumerated below:
 - a) **Identify the major topics or subtopics or concepts** relevant to the subject under consideration.
 - b) **Place the citation of the relevant source** (article/patent/website/data, etc.) **in the correct category of the concept/topic/subtopic**

Example:

1. **Identify topics:** For a study on mental health, you might focus on topics like therapy methods, medication effects, and lifestyle changes.
2. **Sort sources:** You'd categorize research articles about therapy under therapy methods and studies on diet under lifestyle changes.

- While reading, if you **find something important, highlight or mark it** so you can easily come back to it later.
- A complete literature survey should **thoroughly review high-quality work, lay a strong foundation for your research, and show that your work will add something new to the field.**

ANALYSIS AND SYNTHESIS OF PRIOR ART

- After collecting articles for the literature review, **researchers break them down and find useful information.**
- They then **put all this information together to see what conclusions they can draw from the articles as a group.**
- A literature survey grid of N topics and M sources is shown below to help crystalize the information in different categories.

	Source 1	Source 2	...	Source M
Topic 1		✓		
Topic 2	✓			✓
⋮				
⋮				
Topic N	✓	✓		

- A researcher should analyze the relevant information ascertained in below table by undertaking the following steps:
 - **Understanding the hypothesis,**
 - **Understanding the models and the experimental conditions used,**
 - **Making connections (Connecting different pieces of information)**
 - **Comparing and contrasting the various information, and**
 - **Finding out the strong points and the loopholes. (Identify strengths and weaknesses)**
- **If you believe everything in the literature without questioning it, you may struggle to come up with new ideas** and might not carefully check your own research later.
- ***The goal of a literature review is to find gaps or unsolved problems in the research, spot issues with current methods, and suggest new ideas or improvements.***

- No matter where you get your information, you should carefully check each source to make sure it's useful for your research.
- Relying on articles from trusted journals or approved patents can *save time and give you reliable information*.
- Here are *some criteria that can help a researcher evaluate the information*:
 - **Authority**: Who is the author, and what are their qualifications? Who published the information?
 - **Accuracy**: Does the **information seem trustworthy based on what you already know or from reading other sources?** Does the author **include references or a bibliography to back up the information?**
 - **Scope**: Is the information **suitable for your level of understanding or research needs?**

Bibliographic databases

- These are like **organized collections of information** about research articles.
- They **contain details and possibly summaries** of articles from scholarly literature.
- These databases are **useful for gathering information** about citations and research **articles**.
- Researchers can **easily search through these databases** to **find relevant articles** on a **particular topic**.
- Identifying databases quickly helps in exploring ideas or problems efficiently.



1. **PubMed** – Biomedical and life sciences.
2. **Scopus** – Multidisciplinary, with a focus on scientific, technical, and medical content.
3. **Web of Science** – Multidisciplinary, covering sciences, social sciences, arts, and humanities.
4. **Google Scholar** – Multidisciplinary, providing a broad range of academic literature.
5. **IEEE Xplore** – Engineering, electronics, and computer science.
6. **ACM Digital Library** – Computer science and information technology.
7. **JSTOR** – Humanities, social sciences, and arts.
8. **ProQuest** – Multidisciplinary, including theses, dissertations, and newspaper articles.
9. **SpringerLink** – Scientific, technical, and medical research.
10. **PsycINFO** – Psychology and related fields.

11. **ScienceDirect** – Scientific and technical content, mainly in health and physical sciences.
12. **ERIC** (Education Resources Information Center) – Education literature and research.
13. **MEDLINE** – Biomedical sciences, maintained by the U.S. National Library of Medicine.
14. **CINAHL** (Cumulative Index to Nursing and Allied Health Literature) – Nursing and allied health.
15. **EBSCOhost** – Multidisciplinary, with access to various databases in academic and business fields.
16. **Embase** – Biomedical and pharmaceutical research.
17. **Cochrane Library** – Evidence-based health care and medical research.
18. **INSPEC** – Physics, electrical engineering, and computer science.
19. **arXiv** – Open-access preprints in physics, mathematics, computer science, and related fields.
20. **DOAJ** (Directory of Open Access Journals) – Multidisciplinary, focusing on open-access journals.

21. **SSRN** (Social Science Research Network) – Social sciences, including economics, law, and business.
22. **AGRICOLA** – Agriculture and related disciplines.
23. **MathSciNet** – Mathematical literature.
24. **LILACS** – Latin American and Caribbean health sciences literature.
25. **EconLit** – Economic literature and research.
26. **AnthroSource** – Anthropology-related literature.
27. **Sociological Abstracts** – Sociology and related social sciences.
28. **Bibliography of Asian Studies** – Asian studies literature.
29. **PhilPapers** – Philosophy research and scholarly articles.
30. **CAB Abstracts** – Agriculture, environmental science, and related disciplines.

1) Web of Science

- Web of Science (previously known as **ISI or Thomson Reuters**) offers multiple databases and special tools for academic research.
- It's a useful search tool for scholarly materials, **often requiring an institutional license**. Researchers can search for topics by choosing fields like title, author, topic, and more from a drop-down menu.
- The tool also allows sorting results by the number of citations (from highest to lowest) or by publication date.
- To narrow the search, you can use quotes around phrases, add keywords, or use the "**Refine Results**" panel to filter by keywords, material type (like peer-reviewed articles), date, language, and more.
- The "**Cited Reference Search**" option **helps track down articles that have cited an older paper**. This is **helpful for seeing how an idea has been applied, expanded, or improved over time**.

- Using these filters ensures that your search results are relevant, saving time and effort in reviewing useful sources.
- You can adjust the search results based on what you need, either expanding or narrowing the scope using the built-in options on the site.
- Clicking on any search result provides details like the paper's title, authors, journal type, volume, issue number, year, abstract, and keywords. This helps researchers decide if the full paper is worth reading.

Google and GoogleScholar

- Google is a good starting point when you're researching a new topic. It helps you find free information like government reports, organization data, and company documents
- However, it has some drawbacks:
 - It's a **large collection of random information without quality control**, so you **can't always tell where the results are coming from**.
 - The **search options and filters are limited**.
- **Google Scholar** focuses on scholarly literature, but it also has limitations:
 - ❖ **Some results may seem academic but aren't good sources after closer inspection.**
 - ❖ **It doesn't cover everything. Some publishers don't include their content in Google Scholar.**
 - ❖ **It also has limited search tools and refinement options.**

- In **Google Scholar**, you can use **search operators** to make your **searches more precise and find relevant academic literature**. Here's how to use them:

- 1) **OR** – Expands your search by including synonyms or different

Example:

```
climate change OR global warming
```

- 2) **Brackets/Parentheses ()** – Groups similar words with "OR" while combining them with other concepts.

- Example:

```
(climate change OR global warming) AND policy
```

This will return results that mention "climate change" or "global warming" along with "policy."

3) **Quotation marks “ ”** – **Finds exact phrases.**

- Use quotation marks to search for an exact phrase, ensuring the words appear together.

Example:

```
"renewable energy sources"
```

4) **Site:** – Limits your search to a specific domain or website.

- Use **site:** followed by the domain to find results only from that domain.

Example:

```
artificial intelligence site:edu
```

5) **File type:** – Limits the search to specific file types like PDFs, PowerPoint presentations, or Excel files.

Example:

```
machine learning filetype:pdf
```

Effective search: The way forward

- A **scholarly publication** is **written by experts in a specific field** and usually **reviewed by peers** **to ensure accuracy.** These publications **cite all the sources** they use and are intended for other experts or students.
- The **content is often more advanced** than what you'd find in popular magazines.
- While **engineering researchers** mostly rely on articles from **scholarly journals, books, or peer-reviewed sources,** **useful information can also be found in popular publications.** (Magazines, Newspapers, online news websites, blogs, trade magazines, general books and popular science journals)
- **Popular publications** are **less formal, aiming to reach a broader audience, including both experts and beginners,** and **often focus on news and trends** rather than original research findings.

- When searching for information, a researcher must consider **what they need and where to find it**. Not all information is available online, and **it can take time for scholarly work to be published**.
- If **current news is being reported**, **related studies may not exist yet**, so researchers should **look for similar studies or broader information** related to their topic.

Imagine there's a news report about a brand-new technology, say **quantum internet**. Since it's very new, there might not be many scholarly articles or detailed studies on quantum internet itself. In this situation, a researcher should look for **similar or broader studies**, such as:

- Existing research on **quantum computing** (which is related to quantum internet technology).
- Studies on **internet infrastructure** or **network security** (for broader understanding).
- Articles about **quantum communication protocols** that have been researched earlier.

- Searching for literature is a step-by-step process:
 - 1) Try different keywords and search operators;
 - 2) Evaluate results and apply filters;
 - 3) Adjust your search as needed;
 - 4) Check the citations and references in relevant articles.
- Once the search is done, **researchers need to read critically, note key points**, and **compare the findings**. This process may need to be repeated several times. The **end result of the literature review should summarize the important work done and highlight any gaps or challenges in the area of study**. The literature review is an ongoing process throughout the research project.
- **graduate students** (like those in a master's or Ph.D. program) often find it **difficult to understand research papers**, especially if they are full of **complex mathematical concepts or technical details**. However, with practice and regular reading, they will gradually develop the ability to comprehend these papers better.

- It also emphasizes that during a **literature review**, **simply reading papers without actively thinking about how to use the information won't help**. To make progress, **students must engage with the material, identify important insights, and figure out how they can build on it for their own research**.
- As the **student's** understanding of their topic improves, they may **need to go back and review new research or explore new directions in their field**.
- Ph.D. scholars must write a synopsis and conduct an extensive literature review, checking journals and bibliographies.

Introduction to Technical Reading

- **Out of the many research papers available, only a few** will be **directly useful for a researcher's specific work.**
- It's important to know **where to find credible information.** Rely on **peer-reviewed journals and books from trusted publishers** instead of random websites.
- When reading a research paper, **focus on understanding the technical contributions** the authors are making.
- **Reading a research paper isn't like reading a newspaper.** You may need to **read it several times and spend hours** on it to **fully understand** it.
- **Before committing time to read in detail, skim through the paper to see if it's worth your time.** Skim through the introduction, abstract, and keywords to see **if the paper aligns with your research area.**

- **Not every paper requires you to read every detail.** Some sections might not be useful to you, or you might not fully understand certain parts yet. If you're only interested in the results of a study and not the experimental methods, you can skip those sections.
- **Start by reading the title and keywords. If these don't catch your interest, move on to a different paper.** If the title of a paper is about robotics but your research is on software algorithms, it's better to skip it.
- If the **title is interesting, read the abstract** to get a **quick overview** of the paper's content. The abstract should give you an idea of the main findings, and if it doesn't seem relevant to your research, stop reading.
- If the **abstract catches your interest, skip ahead to the conclusion** to see **if the paper's results are relevant to your work.** If a paper on solar panels concludes that a new design improved efficiency, you might want to read further if your work is related to solar energy.
- Look at the **figures, tables, and captions** as they **provide a quick summary of the key results.** A graph showing the improvement of a machine's efficiency over time can tell you more at a glance than several pages of text.

- If the **paper still seems relevant**, go back and **read the introduction** to **understand the background and purpose of the study**. **The introduction** will **explain why the authors choose this specific research** and **how it builds on previous work**.
- **Results and discussion sections are the core of the paper**. Read them carefully to understand the findings and how the authors interpret them. The **results** might **show how a new algorithm performs better than an old one**, and the **discussion will explain why that happens**.
- Only **read sections like “Experimental Setup” or “Modeling”** if you **want to understand the methods in detail**.
- A **researcher must constantly look for relevant papers** and **stay up to date with the field**. For smaller projects, an advisor might suggest key papers, but for larger projects, you'll need to develop your own strategy for finding literature.

Conceptualizing Research

- A good **research objective** should **focus on creating new knowledge** that others in the **research community accept** as important. If you're studying renewable energy, your objective might be to find a new way to improve solar panel efficiency. This new approach should be recognized as valuable by other researchers.
- The **research problem should be original** (not previously studied) and also **solvable with available methods and tools**. Investigating the effects of a new type of biofuel on air quality could be original and solvable if there are tools (like air quality sensors) to measure the effects.
- **Most ideas and methods come from existing studies and knowledge** in the field. It's **essential to review this literature** to inform your research. Before exploring your biofuel idea, you should read previous studies on biofuels and air quality to understand what has already been discovered.

example of **research objectives** for a study on the effects of climate change on agriculture:

Research Objectives:

1. **To assess** the impact of climate change on crop yield in different regions.
2. **To identify** the specific climate factors (e.g., temperature, rainfall) that affect agricultural productivity.
3. **To analyze** how climate change influences the availability of water resources for farming.
4. **To evaluate** the effectiveness of current agricultural practices in adapting to changing climate conditions.
5. **To recommend** sustainable farming practices that can mitigate the negative effects of climate change.

- **To formulate a strong research objective, you need to be well-versed in the existing literature.** This often means you are already at an **advanced level of knowledge in your field**. By the time you decide on your biofuel project, you should have read numerous articles and studies related to both biofuels and air quality.
- At the Ph.D. level, **researchers must consistently read and stay updated on literature** to connect **three key aspects**:
 - > **A significant problem to solve.**
 - > **The existing knowledge that can help solve it.**
 - > **A method for obtaining new knowledge.**

If your research is about biofuels, you need to keep reading recent studies, advancements in biofuel technology, and methods for measuring air quality.

The way these three aspects come together will differ for each researcher and field.
Immersing yourself in literature is crucial to becoming an expert.

While one researcher might focus on biofuel efficiency, another might explore the economic impacts of biofuels. Both need to read relevant literature to refine their objectives.

- **If your research project is smaller or you lack time to become an expert, seek guidance from your supervisor or an experienced researcher.**

If you're a master's student working on a project about a specific type of biofuel, you might consult your supervisor for advice on a suitable research objective based on their expertise.

- **Established researchers can quickly direct you to essential literature, saving you time.**

Your supervisor might recommend key studies on biofuels that provide foundational knowledge, helping you avoid unnecessary reading.

Conceptualizing research involves **creating a clear, original, and solvable research objective** based on existing knowledge.

To succeed, researchers need to be **well-read in their field, seek guidance when necessary, and continuously integrate new information** into their understanding.

CRITICAL AND CREATIVE READING

Critical Reading

- **When reading a research paper, don't just accept what it says. Always question the results and arguments.**

If a paper claims a new method reduces costs, ask, “How did they measure costs? Are there other factors they didn't consider?”

- **Identify the problem.** Consider **if the authors addressed the right issue. Could there be simpler solutions?**

If researchers are working on a complex algorithm to improve efficiency, ask if there are existing simpler methods they overlooked.

- **Limitations** refer to the **factors that may affect the results or conclusions of the research.** These can be **explicitly stated by the authors** or **may need to be inferred by the reader.** “Recognizing these limitations helps you understand the scope of the research”. If the authors don't address these limitations, it could lead to overgeneralizing the findings.

- **Assumptions** are the things that **researchers believe to be true when they do their study, but they don't always say them clearly.** These beliefs are important because they form the base of their work. If these **assumptions turn out to be wrong or incorrect, the results of the research might not be trustworthy or accurate.**
- **Check if the paper has a clear logical flow. Are there gaps in reasoning?** If one section of the paper does not lead logically to the next, it may indicate a flaw in the argument.
- **Be bold in making judgments about the paper, but be willing to change your mind if you find new information.**
- Verify that the data used supports the arguments made and is collected and analyzed correctly.

Creative reading

- Creative reading **requires you to look for new applications or ideas that the authors might have missed**. If a paper discusses a new technology for energy storage, think about how it could be applied in electric vehicles or renewable energy systems.
- Consider if the **findings could be generalized or extended to other areas**.
- when doing research, you should **think about any real-world problems or challenges that could affect the work**. For example, if your research is about developing a new type of material, you should consider how well this material would actually work in real life. Could it last a long time (durability)? Is it too expensive to use (cost-effectiveness)?
- "Future Research Directions" means **identifying what research should be done next based on the findings of a paper**. It involves thinking about **how the research could be expanded or improved**.

- Critical reading involves **questioning and evaluating research**.
- Creative reading requires **an open-minded approach to find new ideas and applications**.

Taking Notes While Reading

1) Reading and Writing Connection:

- Good researchers must read extensively in order to write well.
- **Strong reading skills directly improve writing quality** because **writing relies on the ideas gathered from reading**.
- Example: A researcher who reads many technical papers on AI will be better prepared to write their own paper on the topic because they understand current research trends and methods.

2) Taking Notes Bridges Reading and Writing:

- The act of taking notes while reading helps in transitioning from reading to writing.
- **Notes preserve important details that might be forgotten later**, which aids in writing a clear and well-researched paper.
- Example: After reading a paper on algorithms, a researcher might note down the algorithm's key steps and limitations, which can be useful when writing a review or comparison later.

3) Writing Notes Helps Memory:

- There's a saying that “**the faintest**(refers to text or handwriting that is very light, barely visible, or difficult to read) **writing is better than the best memory**” meaning that written notes are more reliable than trying to remember everything.
- Writing down important points **ensures that critical information isn't lost, even after a long time.**
- Example: A researcher might forget a useful formula from a paper if they don't write it down, but if it's noted, they can easily refer to it later.

4) Highlighting Key Information:

- While reading a paper, **researchers often highlight key definitions, explanations, or concepts** they may need later.
- It's also important to **note down any criticisms or questions** about the paper to address them in future discussions.

5) Summarizing After Reading:

- After thoroughly reading a research paper, it's helpful **to summarize it in a few sentences to capture its main contributions.**
- This **summary** helps in **quickly recalling the paper's essence.**

6) Comparing with Existing Work:

- **To understand a paper's technical contribution**, it's necessary to compare it with other existing works in the same field.
- This comparison highlights whether the **paper presents new ideas, applies existing concepts in new ways, or integrates various ideas into a novel framework.**
- Example: If a researcher finds that a new machine learning paper uses an existing algorithm but applies it to a different industry (like healthcare), they can classify it as an innovative application of known methods.

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Reading Mathematics and Algorithms

1) Mathematics is essential for engineering research:

- Mathematics plays a crucial role in advancing engineering research and practices. It forms the basis of many technical innovations.
- **Example:** In machine learning, mathematical concepts like linear algebra and calculus are necessary for understanding algorithms.

2) Mathematical derivations and proofs are key to research papers:

- Engineering researchers often cannot avoid working with mathematical proofs or derivations. These form the core of technical papers.
- **Example:** When proposing a new algorithm in a paper, mathematical proof is needed to demonstrate its correctness or efficiency.

3) Avoid skipping over important mathematical content:

- Instead of skimming through derivations, researchers should read them thoroughly, especially if the paper seems relevant to their work.

4) Skimming can be done in some cases:

- It is acceptable to skim over sections if they discuss already familiar concepts or content that is too advanced for the current level of research.

5) Advanced or specialized content can be revisited later:

- If a section is too complex or irrelevant at the moment, it can be set aside and revisited when necessary.

6) Challenges with algorithm implementation:

- Implementing complex algorithms in programming languages (like C, C++, or Java) can be prone to errors, even if the researcher is confident in the theoretical aspect.
- Example: A researcher might implement an optimization algorithm in C++ but find that their code doesn't produce the expected results due to subtle programming mistakes.

7) Testing the algorithm by coding it quickly:

- Before spending too much time analyzing the algorithm, researchers might want to code it quickly to verify that it works.

Reading a Datasheet

1) Researchers read field-specific documents:

- Researchers in various engineering fields need to read certain types of documents relevant to their area.

2) Cross-disciplinary need for datasheets:

- Even researchers from other fields may need to work with electronic parts, requiring them to carefully read datasheets.
- **Example:** A mechanical engineer designing a machine might incorporate a sensor, for which they need to refer to the sensor's datasheet.

3) Importance of initial skimming:

- Skimming the datasheet quickly allows a researcher to get a basic idea of what the component does and decide if it's worth reading in more detail.
- Instead of reading every word, they can focus on key points like the component's function, features, and basic specifications.

4) What is a datasheet:

- Datasheets are like instruction manuals for electronic components. They explain what a component does and how to use it in a circuit.

Example: When using a microcontroller in a circuit, a researcher will refer to the datasheet to understand how it processes signals.

5) First page of the datasheet:

- The first page typically contains the component's function, features, and basic specs, along with a functional block diagram.
- Example: The datasheet for a voltage regulator might summarize its input-output voltage range, efficiency, and internal workings in the first page.

6) Pinout details:

- A datasheet provides a "pinout," showing the physical location of each pin on the component, including a marker for pin 1 to ensure proper connection.
- Example: An engineer must consult the pinout when connecting a microcontroller to a circuit to avoid wiring errors.

7) Graphs and performance data:

- Some datasheets include graphs showing the performance of a component under different conditions, like voltage or temperature, and a "safe operating region" to prevent damage.

8) Truth tables and timing diagrams:

- Truth tables describe input-output behavior, and timing diagrams show how data is sent and received from the part.
- Example: A datasheet for a logic gate might include a truth table indicating what output will be generated for different input combinations, while the timing diagram explains how fast data should be processed.

9) Dimensions and PCB layout:

- Datasheets end with the physical dimensions of the part, important for printed circuit board (PCB) design.
- Example: A researcher designing a PCB for an integrated circuit will use the dimensions in the datasheet to ensure the part fits correctly.

10) Shortcut reading to save time:

- When deciding whether to use a new component, careful reading of the datasheet can help researchers avoid mistakes and save time later on.

11) Reading beyond datasheets:

- Engineers need to read not only technical papers and books but also technical documents like datasheets that are specific to their area of research.
- Example: In addition to reading papers on new sensor technology, an electronics engineer needs to read sensor datasheets to understand how to implement them in practical designs.

Chapter -2

Attributions and Citations

Academic writing follows specific rules, especially when it comes to giving credit to other people's ideas, words, or research. These rules include:

- **Citing:** When we use someone else's ideas or words in our own work, we must mention them directly in the text, so readers know where the information comes from.
- **Referencing:** This is listing full details about the sources we cited, usually at the end of our work. It helps readers find the original work for more context.
- **Acknowledgment:** This involves recognizing contributions to the research, often thanking people who helped in specific ways.

Though they're all about giving credit, **citing**, **referencing**, and **acknowledging** are used differently: citing is in-text, referencing is a full list, and acknowledgment is a way to show appreciation for support.

1) Example of citing

- **Citing** is mentioning specific sources in your paper.
- **Example:** If you write, "Studies show that sleep improves memory (Smith, 2022)," you are citing *Smith's* work from 2022 to support your statement.

2) Example of reference

- **Referencing** is the detailed list of these sources at the end.
- **Example:-** Alhomdy, S., Thabit, F., Abdulrazzak, F.H., Haldorai, A., Jagtap, S., 2021. The role of cloud computing technology: a savior to fight the lockdown in COVID 19 crisis, the benefits, characteristics and applications. Int. J. Intelligent Networks 2, 166–174.
<https://doi.org/10.1016/j.ijin.2021.08.001>.

3) Example of acknowledgement

- **Acknowledgement** is recognizing people or organizations that supported your work in other ways.

Example: In a section called *Acknowledgements*, you might write: "I thank Dr. Lee for her valuable feedback on this study and the University Research Fund for supporting this project."

CITATIONS: FUNCTIONS AND ATTRIBUTES

INTRODUCTION

The mammalian world of bats is experiencing a devastation of several species occurring in the eastern portion of the United States because of what has been labeled “White Nose Syndrome” (Blehert et al., 2009). It has been approximately four years since massive die-offs of bats were discovered. The fungus *Geomyces destructans* has been identified and implicated. While most scientists are still uncertain as to whether the fungus is a causal pathogen or merely an opportunistic agent, they are trying to determine how it is spread. Scientists are bewildered as to how to contain WNS. Efforts have taken place by both private and government sectors to restrict cave usage by humans and to do captive colony conservation (CBD, 2010). Scientists are also working on temperature control measures in caves and antifungals that could contain WNS (Vargas, 2009).

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What Citations Do: Citations serve two main purposes:

- **Credit to Others:** Citations acknowledge the original authors of ideas, research findings, or any other contributions used in new work, ensuring they receive recognition.
- **Traceability for Readers:** Citations help readers locate and verify the original sources of information if they wish to examine the source material in detail. This is essential in

When and Where to Use Citations:

- If any part of a new work (like a research paper, patent, or presentation) includes ideas, text, images, sounds, or other material from someone else's work, that source must be cited. This applies to all forms of media, ensuring that others' contributions are clearly acknowledged and plagiarism is avoided.
- **Placement of Citations:** It's usually best to place citations at the end of sentences or paragraphs where the referenced information appears. Citations should contain enough details (author name, title, publication date, etc.) for readers to locate the original source.

- A researcher needs to cite each source twice:
 - (i) in-text citation, in the text of the article exactly where the source is quoted or paraphrased, and
 - (ii) a second time in the references, typically at the end of the chapter or a book or at the end of a research article
- LaTeX, a document preparation system often used by engineering researchers to automatically format documents that comply with standard formatting needs, is very effective to track and update citations

Primary functions of citations:

1) Verification Function: Citations give readers a way to check whether the cited material is used accurately and in the right context. This helps avoid distortion or misrepresentation of information.

In this example, Dr. Lee cites Dr. Smith's study, which shows that spending 30 minutes in urban green spaces daily can reduce stress. By including a citation, Dr. Lee allows readers to verify that she's accurately representing Dr. Smith's findings. This citation lets readers look up Dr. Smith's work to confirm the accuracy and context of the statement. If Dr. Lee had misquoted the study—like saying five minutes instead of 30—readers could identify this discrepancy by checking the original source, preventing misrepresentation and ensuring that information is correctly conveyed.

2) Acknowledgment Function: Citations are a way for researchers to **gain recognition for their contributions**. The **number and quality of citations** a researcher receives can **impact their career**, as they reflect the **researcher's influence and credibility**. Citations are often a **factor in research funding**, as **institutions and grant bodies use them to assess the importance and impact of a researcher's work**.

Example:-

Imagine a researcher, Dr. Jane Doe, publishes a groundbreaking paper on renewable energy storage techniques. Her findings are unique and highly relevant to others in the field, so multiple researchers cite her paper in their own work.

Citations help Dr. Doe **receive recognition** for her work, **increase her professional credibility**, and **improve her access to career and funding opportunities**. The **more her work is cited**, the **more her reputation as a respected researcher grows**, showing her **contributions' importance to advancing knowledge**.

3) Documentation Function: Citations document the development of scientific ideas over time, showing how a particular concept, theory, or technology has evolved. They help establish a historical context for ongoing research.

For instance, in studies on artificial intelligence (AI), researchers often cite foundational works like Alan Turing's 1950 paper on machine intelligence and John McCarthy's contributions to AI programming languages in the 1950s. By referencing these early works alongside recent advancements, researchers can document the evolution of AI concepts and technologies over time. These citations establish a historical context, showing how foundational ideas laid the groundwork for today's AI developments, helping readers understand the progression and buildup of knowledge that supports current research.

Citation Issues and Misuse:

- **Spurious Citations:** Adding citations where they aren't needed or including irrelevant sources just for the sake of filling the reference list is considered inappropriate.
- **Biased Citations:** Favoring citations of work by friends or colleagues without valid connections to the current research can mislead readers and affect objectivity. Conversely, not citing relevant sources due to personal biases (e.g., avoiding certain authors) is also inappropriate.
- **Self-Citations:** Citing one's own previous work is generally acceptable if it is relevant, especially if the current work builds on past research. However, excessive self-citation can be seen as self-promotion.
- **Coercive Citations:** Sometimes, journal editors may pressure authors to include citations to articles in the same journal to artificially inflate the journal's impact factor. This practice is considered unethical because it doesn't serve the reader's needs or the integrity of academic citation.

IMPACT OF TITLES AND KEYWORDS ON CITATION

- **Importance of Titles:** The title of a paper is crucial. It's the first thing people see and **gives a hint about the research topic**. It **helps researchers find relevant papers** when they are looking for information.
- **Title's Role:** A good title not only informs readers **about the paper's content** but also **grabs their attention**. **Some titles are descriptive but not interesting**, while **others are catchy but may not relate to the topic**.
- **Impact on Downloads and Citations:** The title can influence **how many times a paper is downloaded and cited**.
- **Factors Influencing Titles:** Three main factors affect how a title behaves:
 - 1) **Type of Title:** Different styles of titles may have different impacts.
 - 2) **Length of Title:** Longer titles often include more details about the study's methods or results, which can attract more attention and citations.
 - 3) **Specific Markers:** Titles with question marks, colons, or references to certain places may get fewer citations.

- **Use of Keywords:** Including at least two keywords in the title can help more people find and read the paper, leading to more citations.
- Keywords are important as they summarize the main content and topic of the research. They help search engines and databases categorize the research and reach the right audience.

Citing datasets

- The nature of engineering research has evolved rapidly and now relies heavily on data to justify claims and provide experimental evidences and so data citations must fetch proper credit to the creator of the dataset as citations of other objects like research articles.
- Data citations should have provisions to give credit and legal attribution to all contributors, enable identification and access.
- **Identifying who owns a dataset can be challenging, especially with large datasets or when funding issues are involved.**
- **Researchers need to get permission before using data from specific sources.** When **citing a dataset, enough details should be provided** so that someone else can find the same dataset in the future, even if the original link is no longer available.
- It is proper to include a mixture of general and specific information to enable a reader to be certain that the search result is the same dataset that was sought.

Examples:

1. Historical Data, Sotavento (Wind Farm), Corunna, Spain (July 2016): [Accessed: 4 Oct, 2016] Retrieved from <http://www.sotaventogalicia.com/en/real-time-data/historical>
2. Deb, D (2016). [Personnel survey]. Unpublished raw data.

Styles for citations

- Different citation styles organize information about sources (like books, articles, or websites) in various ways because they prioritize different aspects. Some styles focus on making citations short and easy to read, while others emphasize details like the publication date, author names, or source title.

1) ASCE style (American Society of Civil Engineers)

(a)**Reference list**: This part is to be placed in the bibliography or references at the end of the article or report.

Template for books:

Author Surname, Author Initial. (Year Published). Title. Publisher, City, Pages Used.

Example:

Wearstler, K., and Bogart, J. (2004). Modern glamour. Regan Books, NY.

Template for websites:

Author Credentials / Company Name (Year Published). 'Title'. [http://Website URL](http://WebsiteURL) (Oct. 10, 2013).

Example:

Blade cleaning services (2015): <http://www.bladecleaning.com/problematica> (29 Oct, 2016).

Template for journal publications:

Author Surname, Author Initial. (Year Published). 'Title'. Publication Title, Volume number(Issue number), Pages Used.

Example:

Johnston, L. (2014). "How an Inconvenient Truth Expanded The Climate Change Dialogue abd Reignited An Ethical Purpose in The United States". 1-160.

(b) **In-text citation for journals or books:** The following part is to be placed right after the reference to the source of the citation assignment:

Template

(Author Surname/Website URL Year Published)

Examples:

- i. Citation is a very important part of technical writing. (Deb 2016)
- ii. Engineers create devices to monitor mountains so that nearby inhabitants can be warned of impending eruptions. (Teachengineering.org 2014)

2) IEEE style (Institute of Electrical and Electronics Engineers)

IEEE style is standard for all IEEE journals and magazines, and is frequently used for papers and articles in the fields of electrical engineering and computer science. The IEEE style requires endnotes and that references be cited numerically in the text. Those submitting to an IEEE publication should see guidelines for the specific journal or magazine and may also refer to the complete IEEE editorial style manual.

Chapter in an edited book

[1] A. Rezi and M. Allam, “Techniques in array processing by means of transformations,” in *Control and Dynamic Systems*, Vol. 69, Multidimensional Systems, C. T. Leondes, Ed. San Diego: Academic Press, 1995, pp. 133–180.

3) ASME (American Society of Mechanical Engineers)

ASME citation style is widely used in engineering fields, especially for mechanical engineering research.

In-Text Citations:

Example: "The efficiency of the turbine increased by 15% under optimized conditions [1]."

Reference List: At the end of the document, a numbered list of references is provided. Each reference corresponds to an in-text citation number, maintaining the order of appearance.

Formatting of References: ASME has specific rules for each type of source (like books, journal articles, and websites), focusing on elements like authors' initials, titles, and page numbers.

Book:

- Format: [1] Author's Initials. Last Name, Year, *Title of Book*, Edition (if applicable), Publisher, Location.
- Example: [1] J.D. Anderson, 2010, *Fundamentals of Aerodynamics*, 5th ed., McGraw-Hill, New York.

Journal Article:

- Format: [2] Author's Initials. Last Name, Year, "Title of Article," *Journal Name*, Volume(Issue), pp. Page Range.
- Example: [2] A. Kumar and B. Smith, 2022, "Heat Transfer Analysis in Nanofluids," *Journal of Thermal Science*, 45(3), pp. 123-130.

Conference Paper:

- Format: [3] Author's Initials. Last Name, Year, "Title of Paper," in *Proceedings of Conference Name*, Location, Publisher, pp. Page Range.
- Example: [3] T. Li and M. Gupta, 2018, "Advances in Robotics," in *Proceedings of the IEEE International Conference on Robotics*, San Francisco, IEEE, pp. 45-53.

Website:

- Format: [4] Author's Initials. Last Name, Year, "Title of Webpage," Website Name, URL (Access Date).
- Example: [4] R. Chen, 2021, "Renewable Energy Resources," Energy Central, <https://www.energycentral.com> (Accessed October 5, 2023).

Acknowledgments and Attributions

Aspect	Acknowledgments	Attributions
Definition	Recognizes people, organizations, or institutions that supported the research indirectly.	Credits specific intellectual contributions, resources, datasets, or tools created by others.
Purpose	To thank those who provided general support, resources, or funding for the research.	To formally recognize the use of specific ideas, data, software, or methods from other researchers.
Examples of Contributions	Funding agencies, mentors, advisors, lab technicians, administrative staff.	Data sets, software, theories, methods, or frameworks developed by others and used in the study.
Placement in Research	Usually included in a separate "Acknowledgments" section of the paper or thesis.	Typically included in-text or in the reference section where the specific tool, data, or idea is used.
Formality	Expresses general gratitude and is often optional depending on institutional guidelines.	Required when using another's intellectual property, as it upholds research integrity and transparency.

Example Sentence	"We thank Dr. Brown for her advice and the XYZ Foundation for funding."	"This study used the ABC dataset, provided by the National Data Center."
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- In research, "**Acknowledgments and Attributions**" refer to **giving proper credit to individuals, organizations, and resources** that have **contributed to or supported the work**, even if they are not listed as authors on the paper. This is essential for **recognizing the contributions and efforts of others** in a **respectful and transparent way**.
- This section is common in research papers, and it can **appear at the end of the paper** or as a **footnote** if there's no specific rule for where to put it.
- **Types of Support**: Contributions can be classified into six main categories:
 1. **Moral Support**: Encouragement or motivation from others.
 2. **Financial Support**: Funding from agencies or institutions.
 3. **Editorial Support**: Help with writing or editing the research.
 4. **Institutional Support**: Assistance from an organization or university.
 5. **Technical Support**: Providing equipment or technical expertise.
 6. **Conceptual Support**: Help in developing ideas or discussing the project.

Importance of Acknowledgments:

- **Proper Credit:** Even minor contributions should be acknowledged to give proper credit and respect the intellectual property or effort of others.
- **Reward System:** The **"reward triangle"** suggests that **citations, acknowledgments, and authorship** are all **linked and reflect the importance of someone's contribution**.
- In engineering research specifically, **acknowledgments** often go to **technicians, students, funding agencies, or anyone** who provided **significant scientific input, shared data, provided equipment, or contributed to discussions**. Acknowledging these contributions is a sign of respect and appreciation for the collaborative nature of research.

What Should Be Acknowledged?

- Author should acknowledge quotation, ideas, facts, paraphrasing, funding organization, oral discussion or support, laboratory, and computer work.
- (i) **Quotation**: In technical writing such as in the field of engineering, quotes are used very rarely. Quotations are of two types:
 - (a) **Direct quotations**: are used when **author use actual words or sentences in the same order as the original one**. Author should **use quotation marks for the words or sentences** with **proper acknowledgment**.
 - (b) **Indirect quotation**: **summarizes or paraphrases the actual quote**. In such cases, it is important to **acknowledge with proper name and date**.
- Authors should acknowledge people who give appropriate contribution in their research work. **Non-research work contributions are not generally acknowledged in a scientific paper but it may be in a thesis**. Persons must be acknowledged by authors, who gave **a scientific or technical guidance, take part in some discussions, or shared information to author**. Authors should acknowledge **assistants, students, or technicians, who helped experimentally and theoretically** during the research work.

- If the researcher received **grant from a funding agency** and if those funds were used in the work reported in the publication, then such support should always be **acknowledged by providing full details of the funding program and grant number in the acknowledgment section**. The authors should also gratefully **acknowledge use of the services and facilities of any center or organization** with which they are not formally affiliated to.

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- Many technical journals explicitly **discourage authors to thank the reviewers in their article submissions**. This could be construed as **favoritism or an attempt to encourage reviewers to accept their manuscript** for reasons other than scientific merit.

- If the **research results were previously presented at a conference or published as an abstract**, the author should include this information in the acknowledgment.
- **Acknowledgments** show the **author's honesty**, **build credibility**, and **may help others in their careers**.
- **Funding agencies** often require authors to **acknowledge their support**; failure to do so could mean **losing current or future funding**.
- If an **idea, method, or design** is **not common knowledge**, it should be **acknowledged**, even if the review process is anonymous.

Acknowledgments in Books/Dissertations

- An **acknowledgments page** is usually **added at the beginning of a thesis or dissertation, right after the table of contents**. Unlike the brief acknowledgments in journal articles or conference papers, **these are more detailed**. They give the researcher an **opportunity to thank everyone who contributed to completing the research**.
- It's important to think carefully about **whom to acknowledge and in what order**. **Appreciation should be expressed briefly and professionally, without overly emotional language**.
- Typical acknowledgments include the **main supervisor, any secondary supervisors, lab peers, department staff (both academic and technical), colleagues from other departments or institutions, former students, and sometimes family and friends**.

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Thank you all for your contributions to this achievement.

Sample Acknowledgement in Thesis:

I wish to express my sincere appreciation to my supervisor Prof. Gang Tao for the useful comments, remarks and encouragement throughout this thesis work. Furthermore, I wish to express my thanks to Prof. Jacob Hammer for introducing me to the topic and for the support along the way. Also, I like to thank my peers in the Adaptive Control Lab such as Yu Liu and Shan-shan Li, who have shared their precious time during many lively technical discussions. I would like to thank my family members who have supported me throughout this journey in many different ways.

DEDICATION OR ACKNOWLEDGEMENTS

- Dedications are **rarely included in journal papers, conference articles, or patents**. They are **mostly found in longer documents like books, theses, or dissertations**.
- Acknowledgments and dedications serve different purposes: **acknowledgments thank those who contributed to the work** (like helping with editing or providing support), while a **dedication can be to anyone the author chooses, such as a family member, friend, pet, or religious figure**.
- It's possible **to both dedicate a work to someone and acknowledge them**. For example, **an author might dedicate a book to their spouse and also thank them for their support and patience**.

- **Acknowledgments in technical books** are often **short, similar to those in journal articles**. This brevity is because technical books and articles **focus on conveying information efficiently, and acknowledgments are secondary**. The acknowledgment sections usually **recognize contributors or funding sources** without much elaboration.
- The **acknowledgment section of a technical** report (Research report, case studies, project progress report, technical manuals or user guides etc.) may be a **paragraph** that is **longer than a journal paper but shorter than dissertations**.
- Generally, the **length of the acknowledgment** have some **correlation** with the **length of the document**.

Dedication

This book is dedicated to my mentors, friends, and family, whose encouragement and guidance have been a constant source of motivation.